

FarmFusion: Complete System Architecture & Blueprint

Multilingual AI Copilot — Working, AI/ML Models, Datasets & Agent Ecosystem

Target Profile: B.Tech AI/ML Engineering & Production Agritech | Stack: Kotlin Android + FastAPI Async + pgvector + Redis

Executive Summary: FarmFusion is a production-grade, multilingual AI agricultural copilot engineered for Indian smallholder farmers. The system operates across a native Kotlin Android frontend and an asynchronous FastAPI backend (Python 3.13, PostgreSQL 16 + pgvector, Redis). Architecturally, FarmFusion enforces a strict three-tier separation: (1) **Stateful Multi-Turn Agents** for conversational orchestration and telephony; (2) **Deterministic ML Workflows** for high-stakes decisions (crop disease detection, crop recommendation, mandi price forecasting); and (3) **Deterministic Tools** for external ground-truth APIs (live weather, mandi prices, soil moisture, and government schemes). **Core Design Rule: Zero Data Fabrication.** Neural LLMs never estimate weather metrics or invent mandi prices; they only synthesize verified ML/API outputs.

1. System Working & High-Level Architecture

- Frontend Client (Kotlin / Jetpack Compose):** Native Android application using Jetpack Compose, Retrofit HTTP/WebSocket client, and Navigation Component. Communicates via REST APIs on port 8000 and full-duplex WebSockets for real-time telephony voice calls and IoT intrusion push alerts.
- Asynchronous Backend (FastAPI / Python 3.13):** Non-blocking async execution using AsyncIO, SQLAlchemy 2.0 async engine, Pydantic v2 validation, and structlog JSON structured logging. Powers the LangGraph state machine, telephony audio pipes, and ML model workers.
- Hybrid Database (PostgreSQL 16 + pgvector):** Relational tables store farmer profiles, crop histories, price alerts, and IoT events. pgvector HNSW indexing (1024-dimensional) indexes chunked ICAR crop guidelines and central/state government schemes.
- In-Memory Caching & Session Store (Redis):** Ephemeral session checkpoints, 15-minute weather TTL caches, 12-hour mandi forecast caches, and 24-hour multilingual neural audio voice cache keys (tts: {lang}: {hash}).
- Strict Architectural Distinction:** The platform distinguishes between an *Agent* (stateful, dynamic multi-step goal planning), a *Workflow* (fixed-step ML + RAG + LLM pipeline with zero branching ambiguity), and a *Tool* (single-call, deterministic async function).

System Component Classification Matrix

Category	Component Name	Primary Engine / Tech	Role & Execution Boundary
Agent	Main Multilingual Orchestrator	LangGraph StateGraph + Gemma 3 12B	Stateful multi-turn dialogue, intent extraction, slot filling, tool routing.
Agent	Kisan Calling Agent	Telephony WebSocket + STT/TTS + LLM	Full-duplex outbound phone calls for market alerts, weather warnings.
Agent / Tool	Weather Intelligence Agent	Open-Meteo API + TTL Cache	Live/forecast meteorology, soil moisture profiles, smart irrigation schedules.
Workflow	Crop Disease Detection	EfficientNet-B3 (PyTorch) + ICAR RAG	Leaf gatekeeper check, 38-class classification, confidence tiering, treatment.
Workflow	Crop Recommendation	XGBoost 22-Class + Regional Rules	Soil test NPK/pH + historical ERA5 rain + state agronomic re-ranking.
Workflow	Mandi Price Forecaster	Meta Prophet + LightGBM Ensemble	7-day modal price forecast, 95% CI bands, Sell-vs-Wait decision logic.
Tool	Disaster Risk Predictor	4-Model Soft Voting Ensemble	12-feature thermodynamic physical analysis for Drought, Flood, Cyclone.
Tool	RAG Scheme Search	BGE-M3 Embeddings + pgvector HNSW	Vector retrieval of authentic central/state agricultural subsidies.
Hardware Edge	IoT Animal Intrusion	ESP32 C++ + 6 IR + 2 PIR Sensors	Real-time crop perimeter breach detection, deterrence buzzer, push alert.

2. Deep-Dive: Every Agent, Workflow & Tool Breakdown (Part 1)

Each subsystem enforces strict deterministic bounds. Below is the operational specification of components 2.1 through 2.5:

2.1 Main Multilingual Orchestrator (LangGraph Stateful Agent)

Code Path: backend/app/orchestrator/ (graph.py, state.py, nodes/)

Architecture: LangGraph StateGraph with MemorySaver checkpoint for multi-turn session persistence.

Operational Flow: Executes a 3-node directed acyclic graph on every user turn:

- Intent Classification Node:** Analyzes natural language input (voice transcription or text) across 9 canonical intents (weather, mandi_prices, crop_recommendation, disease_detection, disaster_risk, scheme_info, navigation, smalltalk, unknown). Extracts semantic entity slots (crop name, mandi/district, soil NPK values, state). If intent confidence < 0.60 or critical slots are missing, it branches directly to clarification: sets requires_clarification = True and synthesizes a clarifying question without calling tools.
- Tool Router Node:** Dispatches validated intents to asynchronous single-call functions in ToolRegistry. Aggregates structured tool outputs.
- Response Synthesizer Node:** Prompts Gemma 3 12B (fallback: Qwen2.5-7B) via OpenRouter. Enforces strict rural-friendly constraints: short 2–3 sentence responses, dominant language matching (Tier 1/2/3), zero fabricated numbers, and practical agricultural action points.

2.2 Kisan Calling Agent (Telephony Voice Agent)

Code Path: backend/app/calling_agent/ (orchestrator.py, service.py, stt.py, tts.py, prompts.py)

Architecture: Full-duplex WebSocket Telephony Orchestrator (KisanVoiceOrchestrator).

Operational Flow: Manages automated inbound/outbound telephone calls with farmers for critical farm events (severe weather warnings, target mandi price alerts, pest outbreak advisories). Streams raw 8kHz/16kHz PCM audio over WebSockets, executes streaming Voice Activity Detection (VAD) with energy thresholds, runs Telephony STT, streams token completions from Groq / OpenRouter with <800ms latency, and serializes synthesized audio packets via Telephony TTS back to the phone line. Handles real-time farmer barge-in interruption seamlessly.

2.3 Weather Intelligence Agent & Agronomic Advisor

Code Path: backend/app/agents/weather_agent.py & app/tools/weather_tool.py

Operational Flow: Queries Open-Meteo High-Resolution Forecast and Historical Reanalysis APIs. Maintains an in-memory 15-minute TTL cache to guarantee low latency and prevent rate limits. Key modules:

- **Dynamic Reverse-Geocoding:** Automatically maps raw GPS coordinates into recognized district/mandi names across India.
- **Smart Irrigation Advisor:** Retrieves multi-depth soil moisture data (0–1cm, 1–3cm, 3–9cm, 9–27cm) and reference evapotranspiration (ET0) to compute water deficits, explicitly alerting against unnecessary irrigation when heavy rain is forecast.
- **Deterministic Agronomic Rules:** Programmatically evaluates pesticide spray windows (wind < 15 km/h, humidity < 85%, no rain next 6 hrs), frost warnings (temp < 4°C), and crop heat stress conditions.

2.4 Crop Recommendation Workflow (Hybrid ML + Regional Re-ranking)

Code Path: backend/app/workflows/crop_recommendation.py

Operational Flow: Executes a 4-stage deterministic decision pipeline:

- Soil Report Ingestion:** Ingests laboratory Soil Health Card parameters (N, P, K in kg/ha, pH 0–14) via OCR or user confirmation. Never fabricated.
- Climatic Augmentation:** Queries Open-Meteo ERA5-Land for historical cumulative annual rainfall and current temperature/humidity for the farm's location.
- XGBoost Classifier:** Runs the 10-feature gradient-boosted decision tree model to produce softmax probability distributions across 22 crops.
- Regional Validation:** Cross-references ML outputs against state-level ICAR agro-climatic crop suitability databases. Agronomically impossible crops (e.g. coconut in Punjab rabi) are penalized and replaced with the top regionally validated alternative.

2.5 Crop Disease Detection Workflow (Vision CNN + Knowledge Base)

Code Path: backend/app/workflows/disease_workflow.py

Operational Flow: Guards crop leaf diagnostics through a verified multi-layer pipeline:

- Plant Foliage Gatekeeper:** Validates that the uploaded image contains plant tissue before executing neural inference.
- EfficientNet-B3 Inference:** Evaluates 300x300 normalized image tensors across 38 crop disease classes spanning 14 agricultural species.
- Confidence Tier Assignment:** Categorizes output into High (≥ 0.75), Medium (0.45–0.74), Low (0.30–0.44), or Unclear (< 0.30).
- ICAR Knowledge Retrieval:** Retrieves biological controls (e.g. Neem oil, Trichoderma viride) and CIBRC chemical treatments. **Safety Guard:** If confidence is Low or Unclear, chemical fungicide dosages are strictly withheld to avoid toxic misuse.

2. Deep-Dive: Every Agent, Workflow & Tool Breakdown (Part 2)

Operational specification of components 2.6 through 2.10:

2.6 Mandi / Market Intelligence Workflow (Prophet + LightGBM Ensemble)

Code Path: backend/app/ml/market/forecaster.py & app/services/mandi_intelligence.py

Operational Flow: Combines tabular government market data, geodesic geometry, and dual ML algorithms:

- **Geodesic Distance Scoring:** Applies the Haversine trigonometric formula to calculate distance between farmer GPS and regional APMC mandis. Computes a practical composite score: $\text{Score} = 0.50 * \text{Price_Norm} + 0.35 * \text{Distance_Score} + 0.15 * \text{Freshness_Score}$.
- **Dual ML Forecasting Ensemble:** Integrates Meta Prophet (for Fourier annual/weekly seasonal cycles) and LightGBM (for autoregressive lag features) via a 60/40 weighted average to forecast 7-day daily modal prices with 95% confidence intervals.
- **Deterministic Sell vs Wait Advisory:** If expected 7-day price delta > +2.5%, issues 'HOLD'; if < -2.5%, issues 'SELL NOW'; otherwise advises 'STABLE MARKET'. Zero LLM price hallucination.

2.7 Disaster Risk Prediction Tool (4-Model Soft-Voting Ensemble)

Code Path: backend/app/ml/disaster/(inference.py, artifacts/)

Operational Flow: Transforms 5 primary weather readings (temperature, relative humidity, 24-hr rainfall, wind speed, atmospheric pressure) into 12 thermodynamic indices (Wind-Rain Interaction, Rain Intensity, Heat Stress, Atmospheric Instability, Pressure Anomaly). Executes a soft-voting ensemble comprising Random Forest, Gradient Boosting, Extra Trees, and XGBoost (25% weight each). Predicts 4 multi-hazard categories: Cyclone Risk, Drought Risk, Flood Risk, and Low Risk with calibrated probability distributions.

2.8 IoT Perimeter Animal Intrusion Detection Subsystem

Code Path: backend/app/animal_detection/ & esp32/animal_detection/animal_detection.ino

Operational Flow: Protects standing crops from wild animal raiding (boars, nilgai, elephants). An ESP32 microcontroller deployed at farm boundaries monitors 6 Infrared (IR) beam-break sensors and 2 Passive Infrared (PIR) motion sensors. On perimeter breach, the firmware triggers local acoustic/light deterrents and transmits sub-second telemetry over WiFi/HTTP to the backend. The backend persists the state-change in PostgreSQL and broadcasts instant WebSocket push notifications to the farmer's Android device specifying the breached boundary (North, South, East, West).

2.9 Universal Voice & Multilingual Localization Engine

Code Path: backend/app/voice/(provider_router.py, profiles.py, bhashini.py, local/)

Operational Flow: Natively supports 22+ Indian languages across 3 operational tiers:

- **Tier 1 (Full Pipeline):** Hindi, English, Bengali, Gujarati, Marathi, Punjabi, Tamil, Telugu, Kannada, Malayalam.
- **Tier 2 (Partial Pipeline):** Odia, Assamese, Maithili, Santali.
- **Tier 3 (Dialects):** Mewari, Marwari, Bhojpuri, Awadhi, Haryanvi, Rajasthani.
- **Provider Cascading:** Prioritizes genuine local ONNX neural TTS models (e.g. Piper Hindi) when installed on the host. If missing, seamlessly cascades to the Government of India Bhashini API. Implements a 24-hour Redis caching layer for repeated speech phrases to ensure instantaneous playback.

2.10 RAG Knowledge Base & Government Scheme Retrieval

Code Path: backend/app/rag/(embedder.py, retriever.py, ingestion.py)

Operational Flow: Indexes verified agricultural guidelines and welfare schemes (PM-KISAN, PMFBY crop insurance, Kisan Credit Card, PKVY organic farming). Text is ingested, chunked into 512-token segments, and transformed into 1024-dimensional dense vectors using BAAI/BGE-M3. At query time, the retriever executes cosine distance search ($\leq$) against a PostgreSQL pgvector HNSW index. Retrieved chunks are passed to the synthesizer node as strict contextual grounding.

3. AI/ML Models Inventory & Technical Specifications

Complete technical catalog of all artificial intelligence, machine learning, and deep learning models deployed in FarmFusion:

Domain / Model	Architecture & Type	Classes / Horizon	Parameters / Size	Accuracy / Metric	Inference Engine
Crop Disease V2	EfficientNet-B3 Deep CNN (PyTorch / timm)	38 disease classes across 14 crop species	12.2M params (43.6 MB .pth)	99.87% Test Acc 0.9975 Macro F1 99.98% Top-3	PyTorch CUDA / CPU TorchScript
Crop Recommender	XGBoost Multi-Class Classifier (300 trees, depth 6)	22 crop categories (Rice, Maize, Chickpea, etc.)	10 input features (5.03 MB .joblib)	99.55% Test Acc 99.41% 5-fold CV	Scikit-Learn / XGBoost runtime
Disaster (Ensemble)	Soft Voting Ensemble (RF + GB + ET + XGBoost)	4 risk classes (Cyclone, Drought, Flood, Low)	12 physical features (15.65 MB .pkl)	96.71% Benchmark (100% Cyclone, 92% Drought)	Scikit-Learn VotingClassifier
Disaster (Standalone)	Dedicated XGBoost Classifier (200 trees, depth 5)	4 risk classes (Multi-hazard probability)	12 features (1.18 MB .pkl)	96.71% Strict Acc 0.84 Macro F1	XGBoost Native Engine
Mandi Price (Meta)	Facebook / Meta Prophet (Additive Seasonality)	7-day future horizon daily modal price	Fourier order 10 (Dynamic fit)	Mean Absolute Pct Error < 4.8%	CmdStanPy C++ Optimization
Mandi Price (Trees)	LightGBM Regressor with Autoregressive Lags	7-step autoregressive recursive forecasts	100 estimators (Dynamic fit)	Ensemble weight: 40% (Prophet: 60%)	LightGBM C-API library
Conversational LLM	Gemma 3 12B Instruct (Primary via OpenRouter)	Multi-turn reasoning, slot extraction, synthesis	12.0 Billion params	Strict zero-hallucination prompting	OpenRouter / Cloud GPU Inference
Fallback LLM	Qwen2.5-7B-Instruct (Fallback via OpenRouter)	Multi-turn reasoning & regional localization	7.6 Billion params	Automated failover on timeout	OpenRouter API
Dense Embeddings	BAAI / BGE-M3 Dense Multilingual Transformer	Dense document chunks to 1024-dim vectors	567 Million params (2.2 GB weights)	State-of-the-art Indic MTEB benchmark	HuggingFace PyTorch / SentenceTransformers
Speech-to-Text (ASR)	Bhashini ASR Pipeline (MeitY, Gov. of India)	Voice audio to Indian language text	Conformer / Whisper Indic fine-tuned	Real-time rural dialect transcription	ULCA Bhashini Cloud Services
Text-to-Speech (TTS)	Piper ONNX Neural Voice / AI4Bharat Indic-TTS	Indian phoneme-to-waveform audio stream	VITS / FastSpeech2 architecture	Low latency < 200ms; offline capable	ONNX Runtime CPU / C++ Engine

Model Inference Architecture & Execution Notes

- **Deterministic Grounding:** No deep learning or gradient-boosted model output is exposed directly to the user as raw floating-point numbers. Outputs pass through agronomic verification wrappers that contextualize recommendations based on crop season (Kharif, Rabi, Zaid) and state.
- **Model Serialization:** Scikit-learn and XGBoost pipelines are stored in binary format using joblib and pickle, including fitted StandardScaler and LabelEncoder transformers to prevent test-time distribution drift.
- **Confidence Calibration:** The disease model's Brier score is 0.0022 with an Expected Calibration Error (ECE) of 0.0008, ensuring that confidence estimates reliably reflect empirical prediction accuracy.

4. Datasets Used & Data Provenance Audit

Catalog of datasets utilized for training and validating FarmFusion's models:

Subsystem	Dataset Name	Primary Source / Provenance	Sample Count & Splits	Key Features / Target Labels
Crop Disease Detection	PlantVillage + PlantDoc Annotated Dataset	Penn State University (PlantVillage) + IIT Delhi (PlantDoc field conditions)	54,305 total images: - Train: 43,444 (80%) - Val: 5,430 (10%) - Test: 5,431 (10%)	38 disease classes spanning 14 crops: Apple, Blueberry, Cherry, Corn, Grape, Orange, Peach, Bell Pepper, Potato, Raspberry, Soybean, Squash, Strawberry, Tomato.
Crop Recommendation	Indian Soil & Climate Agricultural Dataset	Kaggle Agriculture Repository augmented with ICAR agro-climatic zone reports	2,200 total records: - Train: 1,760 (80%) - Test: 440 (20%) 5-Fold Stratified CV	10 engineered inputs: N, P, K (kg/ha), Temperature (°C), Humidity (%), Soil pH (0-14), Rainfall (mm), NPK_sum, N/P ratio, Temp-Humidity interaction. Target: 22 crops.
Disaster Risk Prediction	Real Indian Historical Disaster Dataset	Open-Meteo ERA5 Reanalysis Archive covering documented Indian disasters (2005–2024)	6,982 genuine records: - Low Risk: 6,527 (93.5%) - Drought: 373 (5.3%) - Flood: 56 (0.8%) - Cyclone: 26 (0.4%)	12 features derived from daily ERA5: temp, humidity, precipitation, wind speed, barometric pressure, rain intensity, heat stress, pressure anomaly, instability. 4 risk targets.
Mandi Market Intelligence	Agmarknet Daily Mandi Bulletin Dataset	Directorate of Marketing & Inspection, Ministry of Agriculture & Farmers Welfare, GOI	2,733 active price records: 145 commodities 268 APMC mandis 16 Indian states	State, District, APMC Market Name, Commodity, Variety, Arrival Date, Min Price (Rs/q), Max Price (Rs/q), Modal Price (Rs/q). Seeded in commodity_price.csv.
Agricultural Schemes (RAG)	Government Schemes & ICAR Manuals	Official portals: pmkisan.gov.in, pmfby.gov.in, ICAR Krishi Vigyan Kendra advisories	Over 1,200 chunks: 512-token segments with 64-token sliding window overlap	Eligibility criteria, subsidy percentages, application procedures, required documentation, CIBRC approved agrochemicals, biological control recipes.
IoT Sensor Breach Logs	Farm Perimeter Hardware Telemetry	Physical ESP32 edge deployment in agricultural field enclosures	Time-series event stream stored in PostgreSQL animal_detections table	Device ID, Sensor ID (IR_1 to IR_6, PIR_1, PIR_2), Digital State (Tripped/Clear), Latency timestamp, Boundary Zone identifier.

5. Non-Negotiable Safety Policies & Architectural Guardrails

- ✓ **Rule 1 — Absolute Weather Ground-Truth:** The LLM is strictly prohibited from estimating, simulating, or generating temperature, rainfall, or wind figures. All weather metrics originate solely from verified Open-Meteo API payloads.
- ✓ **Rule 2 — Mandi Price Integrity:** The LLM never predicts market prices. Price forecasts are generated exclusively by the Prophet + LightGBM mathematical ensemble. The LLM's role is solely to narrate the model output into the farmer's dialect.
- ✓ **Rule 3 — Mandatory Disease Confidence Tiering:** Every diagnostic report must compute and display an explicit confidence tier (**High ≥0.75, Medium 0.45–0.74, Low 0.30–0.44, Unclear <0.30**). If confidence is Low or Unclear, chemical fungicide/pesticide dosage instructions are blocked to prevent toxic chemical misuse.
- ✓ **Rule 4 — Zero Scheme Fabrication:** Government subsidy eligibility criteria must originate exclusively from verified pgvector document chunks. Extrapolation or creative elaboration is blocked at the prompt level.
- ✓ **Rule 5 — Validated Android Navigation:** Dynamic client navigation actions triggered by the voice agent must be strictly validated against the Kotlin app's immutable ALLOWED_DESTINATIONS registry before invocation.

Architectural Audit Completed: All models, agents, workflows, and datasets detailed in this document reflect the exact runtime state of the FarmFusionFinal repository. The codebase is configured for production operation with fully decoupled asynchronous services.